An Exploratory Correlation Test of Luminosity and a Local-Expansion Proxy in Cosmicflows-3 Galaxy Groups

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Abstract

A directional radiation-matter conversion hypothesis predicts a minimal catalog-level sign: after major nuisance trends are controlled, more luminous systems may show larger redshift-distance residuals. We tested this prediction in 858 Cosmicflows-3 galaxy groups using $H_{loc}=v_{CMB}/D$ and total K_s-band luminosity. H_{loc} is an observer-frame proxy derived from redshift and distance; it is not a direct covariant expansion scalar and retains peculiar-motion contamination. A multivariable model adjusted for velocity trend, selection correction, richness, distance quality, and sky direction. The residual correlation was weak ($\rho=0.0850$, $p=0.0128$), but the adjusted slope was positive: $\beta=0.0384$ dex per luminosity dex (95% CI 0.0068-0.0700; $p=0.0176$), equivalent to 9.2% higher H_{loc} per luminosity decade. The strict subset remained positive, whereas the SN Ia-only subset was null. CF3 therefore provides preliminary sign-level consistency evidence that justifies continued testing; it does not establish causation, a specific response function, or a value of κ.

Keywords: local Hubble flow; galaxy-group luminosity; Cosmicflows-3; redshift; peculiar velocity; radiation-matter conversion; correlation test

1. Introduction

Redshift is the appropriate observational starting point for testing expansion-like effects, but it is not a pure expansion coefficient. At finite distance, an observed redshift combines the background Hubble flow with line-of-sight peculiar motion, gravitational and propagation effects, distance uncertainty, and calibration structure [1-4]. Consequently, a redshift-distance ratio can reveal a regional association without directly measuring a covariant local expansion scalar.

A companion phenomenological model proposes that a collective luminous field and its radiation-matter conversion current may contribute an expansion-like regional response [5]. Under an extreme leading-order simplification, the regional closure was written

$$H_D = H_{bg,D} + \sigma_s \chi_{\gamma,s} \frac{G L_{\gamma,D}^{eff}}{c^4 R_D}, (1)$$

where $L_{\gamma,D}^{eff}$ is a conversion-weighted radiation budget and R_D is a regional scale. Equation (1) is an illustrative closure, not the functional form tested here. The underlying fixed-κ contained-volume scaling and event-level impulse picture permit macroscopic accumulation over many events [5,14]; therefore the final luminosity-redshift relation may be nonlinear, history dependent, or scale dependent. The present experiment tests only whether its minimal predicted sign is detectable.

Galaxy groups provide a first catalog-level test because distances, CMB-frame redshift velocities, and aggregate luminosities can be attached to the same systems. Cosmicflows-3 (CF3) contains 17,669 galaxy distance measurements and a summary table of 11,508 group nests [1,6,7]. Group averaging reduces individual orbital-velocity noise but does not remove coherent peculiar flows.

The study asks one deliberately limited question: after controlling major catalog-level trends, is cataloged near-infrared luminosity positively associated with an observer-frame local-expansion proxy? It does not estimate κ, prove a causal expansion mechanism, or determine an analytic response function.

2. Data

2.1 Cosmicflows-3 group catalog

We used CF3 tables 2 and 3 as distributed through VizieR under catalog identifier J/AJ/152/50 [1,8]. Table 2 summarizes 11,508 groups; 1,704 have two or more distance measurements. The fields used were group identifier, weighted mean distance modulus and uncertainty, luminosity distance D, number of contributing distance measurements, group velocity in the CMB frame v_{CMB}, summed K_s-band luminosity $\log_{10}\left(L_{K_s}/L_{\odot}\right)$, luminosity-selection correction factor, group richness, and Galactic sky coordinates.

The near-infrared luminosity originates from the 2MASS group construction [6,7]. It traces the older stellar population more closely than blue-band luminosity, but it is neither bolometric luminosity nor the model's conversion-weighted radiation budget. Its absolute scale also uses a redshift-derived distance convention and a correction for missed light [1]. We therefore analyzed luminosity residuals at controlled velocity rather than treating the published luminosity as statistically independent of redshift.

2.2 Primary and sensitivity samples

The primary sample used the following criteria:

1. $3000 \le v_{CMB} \le 15000$ km s^{-1};
2. at least two group distance measurements;
3. distance-modulus uncertainty $\sigma_\mu \le 0.30$ mag;
4. luminosity correction factor $1 \le C_F \le 10$;
5. finite positive distance, velocity, luminosity, and richness; and
6. $40 \le H_{loc} \le 110$ km s^{-1} Mpc^{-1} to remove catastrophic velocity-distance ratios rather than tune a preferred result.

These filters retained 858 groups spanning $10.06 \le \log_{10}\left(L_{K_s}/L_\odot\right) \le 14.21$. Median H_{loc} was 73.02 km s^{-1} Mpc^{-1}, with an interquartile range of 67.23-79.21.

A strict-quality sensitivity sample required at least three distance measurements, $\sigma_\mu \le 0.20$ mag, and $C_F \le 5$, retaining 242 groups. A separate SN Ia-only sample used median repeated SN Ia distance moduli and retained 188 groups. The SN-only analysis is a sensitivity test, not an independent replication.

3. Methods

3.1 Observer-frame local-expansion proxy

For group i, the proxy was

$$H_{loc,i} = \frac{v_{CMB,i}}{D_i}. (2)$$

For grouped analyses, D_i was the CF3 weighted group distance. For the SN-only analysis, the distance in Mpc was

$$D_{SN,i} = 10^{\left(\mu_{SN,i}-25\right)/5}. (3)$$

Because v_{CMB} is inferred from redshift, H_{loc} is closer to a redshift-distance observable than to a directly measured mechanical velocity. Nevertheless, it remains a mixed proxy. Schematically,

$$H_{loc,i} \simeq H_{bg} + \frac{v_{pec,i} + v_{other,i}}{D_i}, (4)$$

where $v_{pec,i}$ is radial peculiar velocity and $v_{other,i}$ collects additional redshift, propagation, calibration, and any hypothesized conversion-linked residual contributions. Equation (4) explains why group averaging, sky controls, and future independent flow reconstruction are necessary.

3.2 Association model

The primary regression was

$$\log_{10} H_{loc,i} = \alpha + \beta \log_{10} L_{K_s,i} + f_3\left(\log_{10} v_{CMB,i}\right) + Z_i + W_i, (5)$$

where f_3 is a cubic velocity polynomial. Z_i includes the luminosity correction factor, group richness, and number of distance measurements; W_i contains three dipole sky components. HC3 heteroskedasticity-consistent errors, a nonparametric bootstrap, rank correlations, and a stratified permutation test were used. The coefficient β is a conditional association, not a conversion-efficiency estimate. Its percentage interpretation is

$$\Delta H_{loc} = 100\left(10^{\beta} - 1\right)\%. (6)$$

The nuisance controls reduce measured catalog trends but do not reconstruct peculiar velocities or remove unmeasured confounding by mass, environment, morphology, or catalog construction. All tests were two-sided and exploratory.

3.3 Visualization

For Figure 1, primary-sample nuisance-adjusted luminosity residuals were divided into six equal-count bins. Points show back-transformed mean H_{loc} residuals with 95% bootstrap intervals. The straight line shows the positive residual-regression trend; it is a visual summary of the prespecified adjusted model, not a separate fit selected for appearance.

4. Results

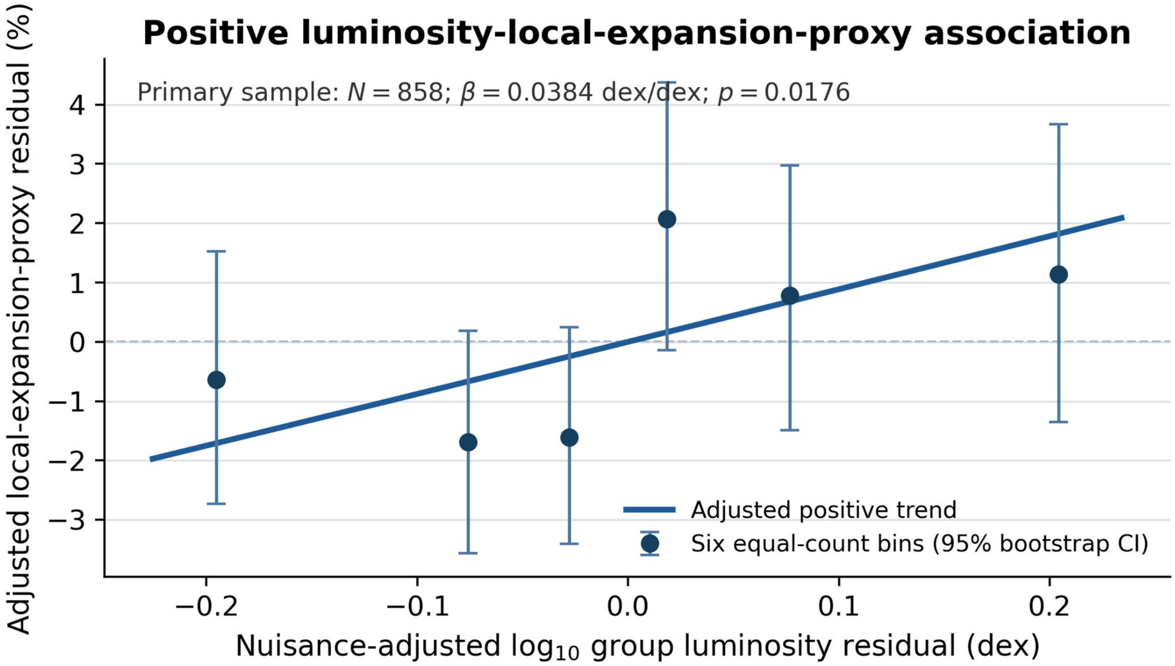
4.1 Primary grouped-distance sample

The unadjusted association was positive but small (Pearson $r = 0.0844$, $p = 0.0134$; Spearman $\rho = 0.0840$, $p = 0.0138$). After controlling velocity and the selection-correction factor in ranks, partial Spearman correlation increased to $\rho = 0.1196$ ($p = 4.46 \times 10^{-4}$).

In the full nuisance-adjusted regression,

$$\beta = 0.0384 \pm 0.0161, 95\% CI = \left[0.0068, 0.0700\right), p = 0.0176. (7)$$

The bootstrap interval was [0.0079, 0.0688]. The point estimate corresponds to 9.2% higher H_{loc} per luminosity decade, conditional on the model. Residual Spearman correlation was $\rho = 0.0850$ ($p = 0.0128$), and stratified permutation gave $p = 0.0144$. Thus, different methods agreed on a weak positive sign.



Primary-sample adjusted positive association

Figure 1. Simplified primary-sample correlation display. Points are six equal-count bins of nuisance-adjusted luminosity and local-expansion-proxy residuals; error bars are 95% bootstrap intervals. The straight line displays the positive adjusted trend. The figure summarizes association only and does not imply causation or a unique functional law.

4.2 Sensitivity samples

The strict-quality sample produced a larger positive adjusted slope, $\beta = 0.0963$ (95% CI 0.0302-0.1624; $p = 0.00468$), equivalent to 24.8% per luminosity decade. Its difference from the primary estimate shows that effect magnitude is selection sensitive.

The SN Ia-only sample was null: $\beta = 0.0106$ (95% CI -0.0159 to 0.0370; $p = 0.435$). It therefore does not confirm the primary association, although its interval remains compatible with a small positive effect near the lower part of the primary interval.

Analysis sample	N	Adjusted β (95% HC3 CI)	Regression p	Residual/permutation p
Primary grouped distances	858	0.0384 (0.0068 to 0.0700)	0.0176	0.0128 / 0.0144
Strict quality	242	0.0963 (0.0302 to 0.1624)	0.00468	0.00157 / 0.00460
SN Ia only	188	0.0106 (-0.0159 to 0.0370)	0.435	0.421 / 0.403

Table 1. Association estimates. β is measured in dex of H_{loc} per dex of cataloged K_s luminosity. All regressions include the nuisance controls in Equation (5).

5. Discussion

5.1 What the positive result supports

The primary CF3 analysis detects a weak positive association between cataloged near-infrared luminosity and a redshift-distance local-expansion proxy after adjustment for major catalog trends. This satisfies the motivating hypothesis's minimal sign-level prediction and provides a reasonable empirical basis for continued testing.

The result does not specify a linear physical law. The observed β is the local elasticity of a noisy catalog proxy, not the causal fraction of expansion attributed to radiation-matter conversion. Event-level momentum-volume scaling may accumulate through many sources, directions, spatial cells, and times; the resulting macroscopic response may therefore require summation, integration, or a nonlinear response function that remains to be derived [5,14].

5.2 Why the result is not proof of local space expansion

Three boundaries are decisive. First, H_{loc} is constructed from redshift and distance but still contains peculiar motion and other propagation or calibration effects; redshift is not itself a pure expansion scalar. Second, cataloged K_s luminosity is redshift-scaled, corrected for missed light, and spectrally incomplete relative to a conversion-weighted luminosity. Third, the SN Ia-only sensitivity sample is statistically null. These limitations allow a positive correlation test but prevent a causal or functional interpretation.

5.3 Relation to the staged theoretical program and a future test

Paper 1 provides a fixed-κ spinor/contained-volume scaling, while Paper 2 provides a simplified conversion-impulse model [5,14]. Under extreme simplifications, microscopic momentum examples, including neutrino-spinor benchmarks, can be compared with macroscopic redshift-luminosity sign tests. CF3 contributes only the sign test; it neither measures κ nor supplies the accumulated analytic response function.

A concise future experiment is to compare periodic variable-star luminosity changes with pulsar-timing-array residuals [15]. A reproducible luminosity-locked signal with a distinguishable scalar **breathing mode** would provide direct phenomenological support for the core space-shift premise [16]. This is a proposed test, not evidence supplied by the present data.

6. Limitations

1. Correlation cannot establish causal expansion by luminosity.
2. H_{loc} is affected by peculiar velocity and is not a direct local metric-expansion measurement.

3. CF3 combines distance indicators with different selection functions and calibration systematics.
4. Cataloged K_s luminosity is redshift-scaled and is not the model's bolometric conversion-weighted radiation budget.
5. The primary and strict samples are nested, so their agreement is not independent replication.
6. The SN Ia-only sample does not reproduce the main association.
7. Mass, environment, velocity dispersion, morphology, and catalog construction remain possible confounders.
8. No precise analytic luminosity-expansion function or numerical κ prediction is tested here.

7. Conclusions

Among 858 CF3 galaxy groups, cataloged K_s luminosity is weakly but positively associated with the observer-frame proxy $H_{loc}=v_{CMB}/D$ after major nuisance adjustment. The primary estimate corresponds to approximately 9.2% higher H_{loc} per luminosity decade, and the strict-quality sample remains positive; the SN Ia-only sample is null.

CF3 therefore provides preliminary sign-level consistency evidence that makes continued investigation of the radiation-matter conversion hypothesis reasonable. It does not prove that luminosity expands space, establish a particular mathematical function, or determine κ. The appropriate next steps are independent luminosity and distance calibration, peculiar-flow reconstruction, derivation of the accumulated response function, and targeted future tests such as a luminosity-linked PTA breathing-mode search.

Declarations

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Conflicts of Interest. The author declares no conflict of interest.

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